

Cryptography

Cryptography is the science of protecting information by converting it into a secure form (ciphertext) so that only authorized users can access or understand it. It uses mathematical algorithms and keys for encryption and decryption to ensure confidentiality, integrity, authentication, and non-repudiation.

Cyber Security

Cyber security is the practice of protecting computers, networks, systems, and data from cyber attacks, unauthorized access, damage, theft, and disruption. It involves technologies, processes, and policies to ensure the confidentiality, integrity, and availability (CIA) of information.

Classical Cryptography

Classical cryptography refers to traditional encryption techniques that were used before the arrival of computers. These methods mainly rely on substitution and transposition of characters.

Some key Characteristics of Classical cryptography:

- ✓ Uses simple mathematical techniques.
- ✓ Same key is often used for encryption and decryption.
- ✓ Easy to implement but relatively easy to break with modern computing.

Examples: Caesar Cipher, Vigenère Cipher, Playfair Cipher, Hill Cipher, Rail Fence Cipher

Modern Cryptography

Modern cryptography uses advanced mathematical algorithms and computer-based techniques to provide strong security for digital communication and data storage.

Some Characteristics of Modern Cryptography:

- ▶ Uses complex mathematical algorithms.
- ▶ Supports both symmetric-key and asymmetric-key encryption.
- ▶ Provides confidentiality, integrity, authentication, and non-repudiation.
- ▶ Highly secure against modern attacks.

Examples: AES (Advanced Encryption Standard), DES and 3DES, RSA, ECC (Elliptic Curve Cryptography), SHA-256 (Hash Function)

Difference Between Classical and Modern Cryptography

Classical Cryptography	Modern Cryptography
Uses substitution and transposition techniques.	Uses advanced mathematical algorithms.
Designed for manual encryption.	Designed for computer-based encryption.
Less secure and vulnerable to attacks.	Highly secure and resistant to attacks.
Mainly uses symmetric encryption.	Uses both symmetric and asymmetric encryption.
Examples: Caesar, Vigenère, Playfair, Hill.	Examples: AES, RSA, ECC, DES.

Modular Arithmetic in Cryptography

Definition: Modular arithmetic is a system of arithmetic in which numbers "wrap around" after reaching a certain value called the modulus (n). It is also known as clock arithmetic and is widely used in cryptographic algorithms such as RSA, Diffie–Hellman, and ECC.

Modular Operation

The notation is:

$$a \equiv b \pmod{n}$$

This means that a and b have the same remainder when divided by n , or equivalently, n divides $(a - b)$.

When two numbers a and b leave the same remainder when divided by n , we can express this mathematically as $a \equiv b \pmod{n}$ (Read as: a is congruent to b modulo n)

Let $a = 17$, $b = 12$, and $n = 5$.

- Dividing 17 by 5: $17 = 3 \cdot 5 + 2$ (Remainder is 2)
- Dividing 12 by 5: $12 = 2 \cdot 5 + 2$ (Remainder is 2)

Since both numbers yield the same remainder (**2**), we can write:

$$17 \equiv 12 \pmod{5}$$

Let $a = -2$, $b = 8$, and $n = 10$.

- Dividing -2 by 10: $-2 = (-1) \cdot 10 + 8$ (Remainder is 8)
- Dividing 8 by 10: $8 = 0 \cdot 10 + 8$ (Remainder is 8)

Since both numbers yield the same remainder (**8**), we can write:

$$-2 \equiv 8 \pmod{10}$$

Importance in Cryptography

Modular arithmetic is the foundation of many cryptographic algorithms because it:

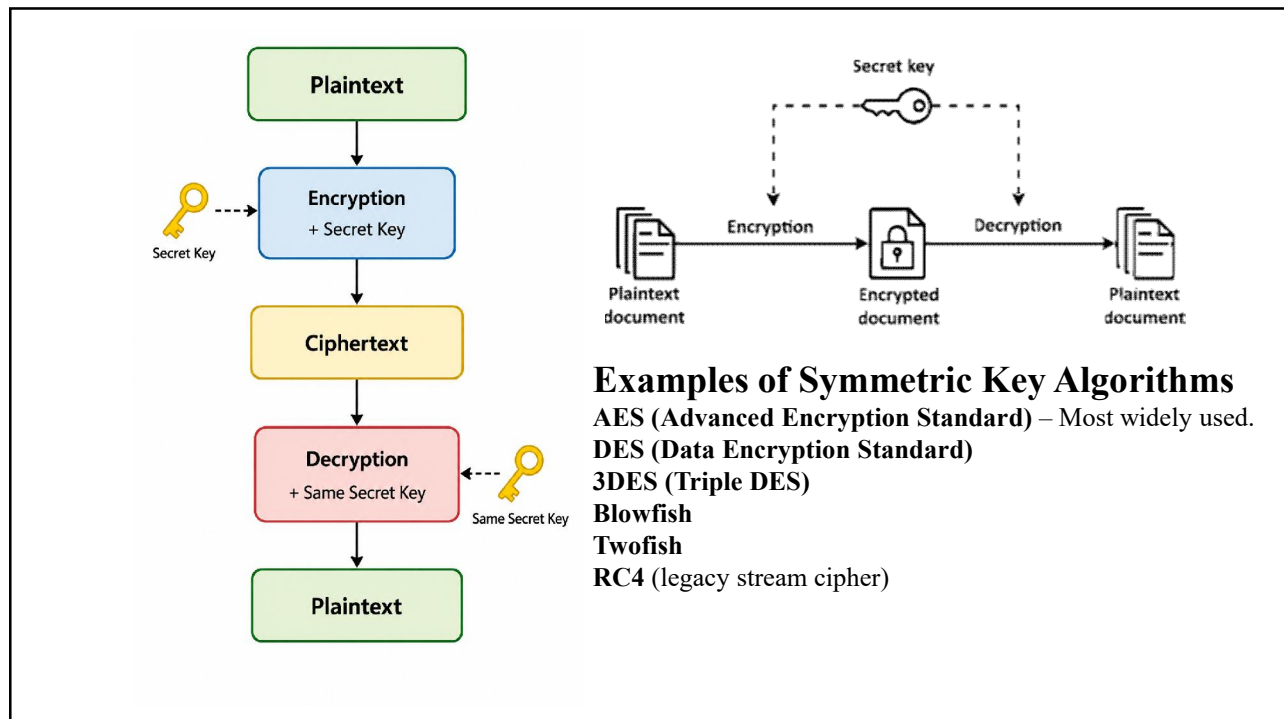
- ▶ Keeps numbers within a fixed range.
- ▶ Makes encryption and decryption efficient.
- ▶ Provides strong mathematical security.
- ▶ Enables public-key cryptography.

Symmetric Key Encryption

Symmetric key encryption is a cryptographic technique in which the same secret key is used for both encryption (converting plaintext into ciphertext) and decryption (converting ciphertext back into plaintext).

Working Process

1. The sender encrypts the plaintext using a secret key.
2. The encrypted data (ciphertext) is sent to the receiver.
3. The receiver uses the same secret key to decrypt the ciphertext and recover the original plaintext.



Caesar Cipher

The Caesar Cipher is one of the oldest and simplest methods of encrypting a message. Named after Julius Caesar, who used it to protect his military communications, it is a type of substitution cipher.

In this cipher, each letter in the original message (plaintext) is shifted down the alphabet by a fixed number of positions.

How It Works

Imagine the alphabet wrapped in a circle. If you choose a shift (often called the **key**), you move every letter forward by that exact number of steps.

For a standard shift of **3** (the traditional choice used by Caesar):

Plaintext Alphabet: A B C D E F G H I J K L M N O P Q R S T U V W X Y Z

Ciphertext Alphabet: D E F G H I J K L M N O P Q R S T U V W X Y Z A B C

Caesar Cipher

Caesar Cipher is a substitution cipher in which each letter in the plaintext is replaced by the letter that is k positions down the alphabet.

Working ($K = 3$)

Plain Alphabet	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Cipher Alphabet	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C

Encryption

Plaintext	:	H	E	L	L	O
		+3	+3	+3	+3	+3
		↓	↓	↓	↓	↓
Ciphertext	:	K	H	O	O	R

Decryption

Ciphertext	:	K	H	O	O	R
		-3	-3	-3	-3	-3
		↓	↓	↓	↓	↓
Plaintext	:	H	E	L	L	O

Example

Plaintext	:	A	T	T	A	C	K	A	T	D	A	W	N
		+3	+3	+3	+3	+3	+3	+3	+3	+3	+3	+3	+3
		↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓
Ciphertext	:	D	W	W	D	F	N	D	W	G	D	Z	Q

General Formula

Encryption : $C = (P + K) \bmod 26$

Decryption : $P = (C - K) \bmod 26$

P = Position of plaintext letter
 C = Position of ciphertext letter
 K = Key (shift value)
 $(A = 0, B = 1, \dots, Z = 25)$

Encryption Example:

If we want to encrypt the word "HELLO" with a shift of 3:

H shifts 3 places forward -----> **K**

E shifts 3 places forward -----> **H**

L shifts 3 places forward -----> **O**

L shifts 3 places forward -----> **O**

O shifts 3 places forward -----> **R**

Plaintext: HELLO

Ciphertext: KHOOR

To **decrypt** the message, you simply reverse the process by shifting backward by the same key value (in this case, 3).

Mathematical Formula

We can express the Caesar Cipher using the modular arithmetic concepts we discussed earlier.

First, assign each letter a number from 0 to 25 ($A = 0, B = 1, \dots, Z = 25$).

Let p be the numerical value of a plaintext letter, c be the value of the ciphertext letter, and k be the key (shift).

Encryption Formula:

$$c \equiv (p + k) \bmod(26)$$

Decryption Formula:

$$p \equiv (c - k) \bmod(26)$$

Why Modulo 26? The $\bmod\{26\}$ ensures that if a shift goes past 'Z' (index 25), it wraps naturally back around to the beginning of the alphabet ('A').

For example, if $p = 24$ ('Y') and $k = 3$, then

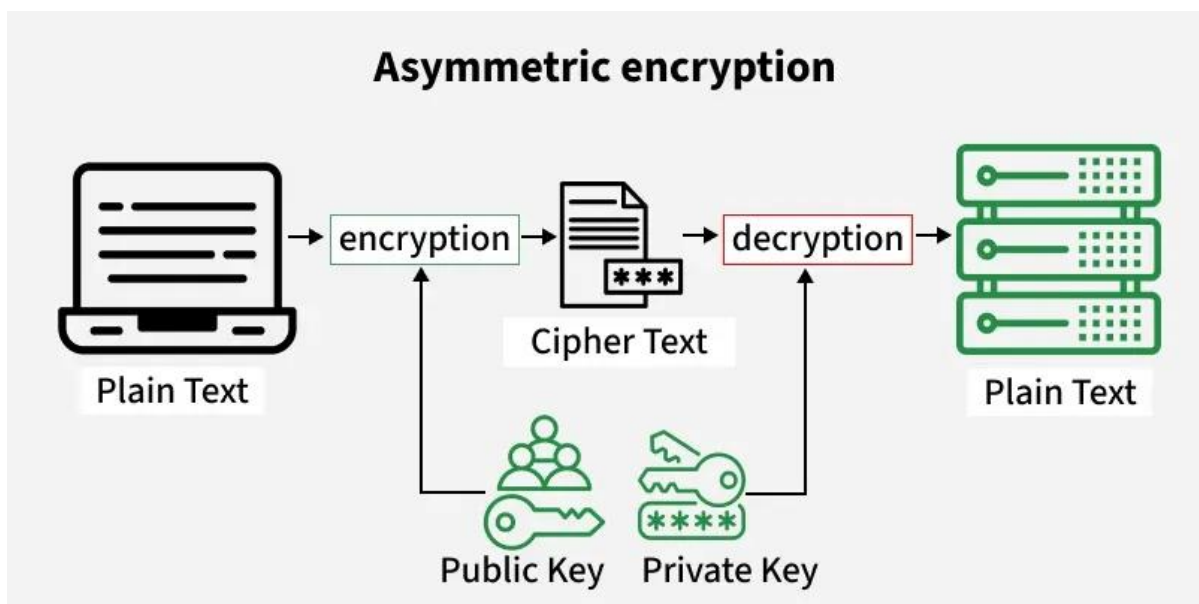
$$\begin{aligned} &(24 + 3) \bmod\{26\} \\ &= 27 \bmod\{26\} = 1 \text{ ('B')}. \end{aligned}$$

Public Key Encryption (Asymmetric Encryption)

Public Key Encryption (also called **Asymmetric Encryption**) is a cryptographic technique that uses **two different keys**:

Public Key – Used to encrypt data and can be shared with everyone.

Private Key – Used to decrypt data and must be kept secret by the owner.



RSA Cryptosystem

RSA is one of the most widely used public-key (asymmetric) encryption algorithms. It was developed in 1977 by **Ron Rivest**, **Adi Shamir**, and **Leonard Adleman**. The name RSA comes from the initials of its inventors.

RSA is used for:

- ▶ Secure data encryption
- ▶ Digital signatures
- ▶ Secure key exchange

RSA uses two keys:

Public Key (e, n) – Used for encryption.

Private Key (d, n) – Used for decryption.

Digital Signature

A Digital Signature is a cryptographic technique used to verify the authenticity, integrity, and non-repudiation of a digital message or document. It is created using the sender's private key and verified using the sender's public key.

Hash Function

A Hash Function is a mathematical function that converts an input message of any length into a fixed-length output, called a hash value, hash code, or message digest.

The hash value is unique for the input, and even a small change in the message produces a completely different hash.

Step 1: Create Message Digest

The sender applies a **hash function** (e.g., SHA-256) to the message.

Message $\longrightarrow$ Hash Function $\longrightarrow$ Digest

Step 2: Generate Signature

The sender encrypts the message digest using **their private key**.

$$\text{Signature} = (\text{Digest})^d \bmod n$$

where d is the sender's private key.